

Целые точки и выпуклые тела

Homework

1 сентября 2026 года

1. (classical) Prove that among any five points of $\mathbb{Z}^2$ two have their midpoint in $\mathbb{Z}^2$, and that the same holds for any $2^n + 1$ points of $\mathbb{Z}^n$; show that $2^n + 1$ is optimal.
2. Prove Minkowski's theorem for a general lattice and deduce Hermite's bound: the shortest nonzero vector of Λ has length at most $2\omega_n^{-1/n}(\det \Lambda)^{1/n}$.
3. (Dirichlet) Deduce from Minkowski's theorem that for irrational α there are infinitely many rationals p/q with $|\alpha - p/q| < 1/q^2$; state the simultaneous version.
4. State Minkowski's second theorem on successive minima and prove it for $n = 1$ and $n = 2$.
5. (MIPT 2021) Find the least c_n such that for every closed symmetric convex body $K \subseteq \mathbb{R}^n$ containing exactly m points of $\mathbb{Z}^n$, every translate of K contains at most $c_n m$ points of $\mathbb{Z}^n$.
6. (RMO) Let a_n be the number of lattice points in the closed disc of radius n centred at the origin. Determine $\lim_{n \rightarrow \infty} a_n/n^2$ and estimate the error term.
7. Compute the Ehrhart polynomial $L_P(t) = |tP \cap \mathbb{Z}^n|$ for the unit square, the triangle $\text{conv}\{0, e_1, e_2\}$ and the standard simplex in $\mathbb{R}^3$; state Ehrhart reciprocity and verify it on these examples.
8. Prove the flatness theorem in the plane: there is c such that every planar convex body without interior lattice points has lattice width at most c ; give a body with lattice width at least 2.

9. (MIPT 2012) Let $X \subseteq \mathbb{Z}^2$ be finite with more than one element, and suppose integer translates of X tile $\mathbb{Z}^2$. Prove that $P(x, y) = \sum_{(n, m) \in X} e^{i(nx + my)}$ vanishes somewhere on $\mathbb{R}^2$.
10. (Minkowski–Hlawka) Prove by averaging over lattices that for every n there is a lattice of determinant 1 whose shortest vector has length at least $c\sqrt{n}$, and compare with Problem 2.